

FARM2HOME

CHAPTER 1

Introduction

Farm2Home is an e-commerce application that connects farmers directly with customers through an online platform. The main purpose of this application is to provide fresh, high-quality agricultural products such as vegetables, fruits, dairy products, grains, and other farm produce at affordable prices.

In the traditional supply chain, agricultural products pass through several middlemen before reaching customers, which increases prices and reduces farmers' profits. Farm2Home solves this problem by enabling farmers to sell their products directly to consumers. Customers can easily browse products, place orders, make online payments, and receive fresh products at their doorstep.

The application benefits both farmers and customers by ensuring fair prices, fresh products, and a convenient shopping experience. It also supports digital agriculture and promotes sustainable farming by reducing unnecessary intermediaries. Farm2Home is designed to provide a secure, user-friendly, and efficient online marketplace for buying and selling farm products.

Objective

The main objective of the Farm2Home application is to connect farmers directly with customers through an online platform. It aims to provide fresh and high-quality farm products at reasonable prices while helping farmers earn better profits by reducing the role of middlemen. The application also offers a simple, secure, and convenient way for customers to purchase farm products from their homes.

Key Objectives of Farm2Home

- Connect farmers directly with customers.

- Provide fresh and high-quality farm products.
- Reduce middlemen and ensure fair prices.
- Help farmers earn better profits.
- Offer easy online shopping and home delivery.
- Provide a secure and user-friendly platform.
- Save customers' time and effort.
- Support local farmers and promote digital agriculture.

Motivation

The motivation behind Farm2Home is to solve the problems faced by both farmers and customers in the traditional agricultural market. Farmers often receive low prices because of intermediaries, while customers pay higher prices for the same products. Farm2Home creates a direct connection between farmers and consumers, ensuring fair prices, fresh products, and an easy online shopping experience.

Stakeholders

The main stakeholders of the Farm2Home system are:

- Customers – Browse, order, and purchase fresh farm products.
- Farmers – Add, manage, and sell their farm products.
- Administrator – Manages users, products, orders, payments, and the overall system.
- Delivery Partners – Deliver products from farmers to customers.

Functional Requirements

1. User registration and login.

2. Farmer registration and login.
3. Browse and search products.
4. View product details.
5. Add products to the shopping cart.
6. Place orders and make payments.
7. Track order status.
8. Farmers can add, edit, and remove products.
9. Customers can view order history and provide ratings and reviews.
10. Admin can manage users, products, orders, and payments.

Non-Functional Requirements for Farm2Home App

1. Performance

- The app should load the home screen within 3 seconds under normal network conditions.
- Support at least 500 concurrent users without performance degradation.
- Order placement should complete within 5 seconds after confirmation.

2. Scalability

- The system must handle growth in users, products, and farms without major redesign.
- Cloud infrastructure should allow horizontal scaling to meet peak demand (e.g., festive seasons).

3. Availability & Reliability

- Ensure 99.9% uptime for critical services (ordering, payment, delivery tracking).
- Automatic failover and backup mechanisms to prevent downtime.

4. Security

- All user data (personal + payment) must be encrypted using AES-256.

- Implement two-factor authentication for farmers and admins.
- Follow GDPR and Indian IT Act compliance for data privacy.

5. Usability

- Simple, intuitive UI for farmers and customers with minimal learning curve.
- Support multi-language interface (e.g., English + Tamil).
- Accessibility features: readable fonts, voice assistance, and screen reader compatibility.

6. Maintainability

- Codebase should follow modular architecture for easy updates.
- Clear documentation for APIs and backend services.
- Automated testing pipeline to ensure smooth deployment.

7. Portability

- App should run on Android (min v8.0) and iOS (min v13).
- Web version should be compatible with major browsers (Chrome, Edge, Safari).

8. Data Integrity

- Ensure no duplicate or missing records during transactions.
- Maintain audit logs for all critical operations (orders, payments, deliveries).

9. Supportability

- Provide 24/7 customer support via chatbot and helpline.
- Farmers should have a dedicated support channel for onboarding and issue resolution.

Problem Statement

In the traditional agricultural market, farmers often depend on middlemen to sell their products. As a result, farmers receive lower profits, while customers pay higher prices for fresh produce. There is also a lack of transparency in pricing,

limited access to fresh farm products, and inconvenience in purchasing directly from farmers.

The Farm2Home application addresses these problems by providing an online e-commerce platform where farmers can sell their products directly to customers. This system helps farmers earn better income, allows customers to buy fresh and quality products at fair prices, and offers a simple, secure, and convenient shopping experience with home delivery.